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An Evaluation of Bagging, Boosting, and Random Forests for Land-Cover Classification in Cape Cod, Massachusetts, USA

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Abstract: The iterative and convergent nature of ensemble learning algorithms provides potential for improving classification of complex landscapes. This study performs land-cover classification in a heterogeneous Massachusetts landscape by comparing three ensemble learning techniques (bagging, boosting, and random forests) and a non-ensemble learning algorithm (classification trees) using multiple criteria related to algorithm and training data characteristics. The ensemble learning algorithms had comparably high accuracy (Kappa range: 0.76–0.78), which was 11% higher than that of classification trees. Ensemble learning techniques were not influenced by calibration data variability, were robust to one-fifth calibration data noise, and insensitive to a 50% reduction in calibration data size.

INTRODUCTION

Land monitoring programs rely upon methods of digital map production that integrate image processing techniques, ground reference observations, and satellite data (Gao, 2002; Hansen et al., 2011; Rodríguez-Galiano et al., 2012b). The integrative nature of the process of land-cover mapping makes it difficult to obtain consistent land-cover accuracy because of complex measurement space as well as calibration data error/noise and variability (Cihlar, 2000; DeFries and Chan, 2000; Zhang et al., 2010). Land-cover mapping in heterogeneous landscapes is challenging because of scene variability introduced by differences in environmental gradients (e.g., moisture, elevation, and temperature), seasonal variability, topographical heterogeneity, and artifacts caused by both ongoing and legacy disturbances (Franklin and Wulder,

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2002; Rogan and Miller, 2007). Heterogeneous landscapes are characterized by land-cover categories that are difficult to reliably discriminate without substantial on-screen editing by analysts due to low inter-class separability and high intra-class variability (Ghimire et al., 2010).

Another major challenge in large area land-cover mapping is minimizing the task of the analyst by utilizing algorithms that have high levels of operability (i.e., ease of use) and automation, as well as readily interpretable classification rules (DeFries and Chan, 2000; Rogan et al., 2008). Moreover, in many instances, the collection of sufficient calibration samples to represent diverse land-cover types in heterogeneous regions is constrained by analyst experience, time, and cost; therefore, proper algorithms are required to produce acceptable map accuracy with as few calibration samples as possible (Rogan and Chen, 2004). The quality of calibration data can be compromised and inadequate for operational purposes due to error introduced by the analyst (Woodcock and Gopal, 2000), global positioning system (GPS) locational uncertainty (McRoberts, 2010), and other issues such as geometric inconsistency or scale issues (e.g., pixel size mismatched with the calibration samples dimension) (Lunetta et al., 1991), and radiometric error (Hill and Sturm, 1991).

Previous studies have used different classification algorithms ranging from parametric to non-parametric approaches to map land cover (Rogan and Chen, 2004). Parametric approaches (e.g., maximum likelihood) that assume data to be normally distributed and require high numbers of calibration sites have been frequently employed for land-cover characterization (Strahler, 1980; Townshend et al., 1987; DeFries and Townshend, 1994; Lu et al., 2011) but are problematic because remotely sensed and ancillary data used in heterogeneous landscapes can be non-normal, multi-model, and categorical. Non-parametric approaches that do not assume a particular data distribution have gained heightened research interest in the last decade (Warner et al., 2009; Rodriguez-Galiano et al., 2012a). Classification trees (Breiman et al., 1984; Hansen et al., 1996; Friedl and Brodley, 1997), neural networks (Foody, 1995; Atkinson and Tatnall, 1997) and support vector machines (Huang et al., 2002; Foody and Mathur, 2004) are some of the more popular non-parametric data driven models used to map land-cover. However, these non-parametric methods tend to over-fit the calibration data, and are often difficult to operate because of the trial and error required in determining user-defined parameter values for algorithm calibration (Foody, 1995; Friedl and Brodley, 1997; Huang et al., 2002; Mountrakis et al., 2010).

Ensemble learning algorithms could benefit complex mapping scenarios because the iterative nature of these algorithms produces multiple classifications, which ensures a convergence approach to pixel labeling and considerably improves classification accuracy because the error of a single classification is outweighed by the ensemble of multiple classifications (Friedl et al., 1999; Gislason et al., 2006; Ghimire et al., 2010; Sesnie et al., 2010; Rodriguez-Galiano et al., 2012b). Ensemble learning algorithms generate a large set of classifications from calibration data using a base learning algorithm (e.g., classification tree) (Friedl et al., 1999; Breiman, 2001; Chandra and Yao, 2006; Zhang et al., 2010; Galar et al., 2011; Rodriguez-Galiano et al., 2012b). Subsequently, the collection of multiple classifications is combined using a rule-based approach (e.g., majority voting) which assigns a pixel to a class having the maximum number of votes from individual classifiers (Breiman, 2001; Kotsiantis and Pintelas, 2004).

Previous studies mostly concur that the three popular ensemble learning techniques (namely, bagging, boosting, and random forests) have comparable overall accuracy (DeFries and Chan, 2000; Gislason et al., 2006), which tends to be higher than the accuracy of a single classification tree (DeFries and Chan, 2000; Gislason et al., 2006; Na et al., 2010; Rodriguez-Galiano et al., 2012b). Studies have also reported that ensemble learning algorithms have similar accuracy compared to more complex machine learning algorithms, such as support vector machines (Pal, 2005; Sesnie et al., 2010). In contrast, other comparison studies have demonstrated that ensemble learning techniques, such as random forests and bagging, outperform computationally demanding non-ensemble learning techniques, such as support vector machines, and neural networks due to the characteristics of the calibration data (e.g., absence/presence) and predictor variables (e.g., physiological environmental variables) employed in the classification (Pino-Mejías et al., 2010). However, few previous studies in land-cover classification have used multiple criteria related to algorithm and calibration data characteristics to evaluate ensemble learning algorithms (DeFries and Chan, 2000; Chan and Paelinckx, 2008; Rodriguez-Galiano et al., 2012b). Multiple criteria evaluations of ensemble learning algorithms have either focused on small geographic regions (area < 100 km²) (Chan and Paelinckx, 2008) or assessed the performance of only one or two ensemble classifiers (DeFries and Chan, 2000; Chan and Paelinckx, 2008; Rodriguez-Galiano et al., 2012b). Thus, there is a need for an extensive multi-criteria evaluation of ensemble learning algorithms for land-cover mapping, especially to understand the characteristics of machine learning algorithms that enable them to perform better than other algorithms, and to understand site-specific differences in algorithm performance.

This study evaluates three ensemble learning algorithms (bagging, boosting, and random forests) and one non-ensemble learning technique (classification tree) to map land cover in Cape Cod, Massachusetts, USA. The questions addressed in this research are: How sensitive are ensemble learning algorithms to the user-defined parameters (e.g., number of trees) used to operate the algorithm? How do ensemble learning algorithms perform with variations in calibration noise? Do ensemble learning algorithms perform well with relatively small sample sizes?

STUDY AREA

The study area is Cape Cod, which is the easternmost region in the state of Massachusetts (Fig. 1). Cape Cod is a complex landscape in a remote sensing context due to variations in environmental conditions caused by gradients in elevation, topography, and soils, and is undergoing rapid land-cover change and urban development with a sharp reduction in the area of undeveloped land (DeNormandie and Corcoran, 2009; Motzkin et al., 2002). Cape Cod is particularly heterogeneous from a remote sensing perspective because of bright soils, peri-urban development, and limited footprint of key land-cover classes related to small area. Physical features are related to glacial origin and there are five major surficial geology types, namely outwash, moraine, beach and dune, ice contact deposits, and lake and swamp deposits (Oldale et al., 1995). Elevation ranges from 60 m at Buzzards Bay and Sandwich moraines in the northwest and north to less than 30 m in the remainder of the study area, which is outwash and mostly flat (Parshall et al., 2003). The dominant vegetation type in the region is pitch pine and scrub oak (Hall et al., 2002). The region has a humid continental

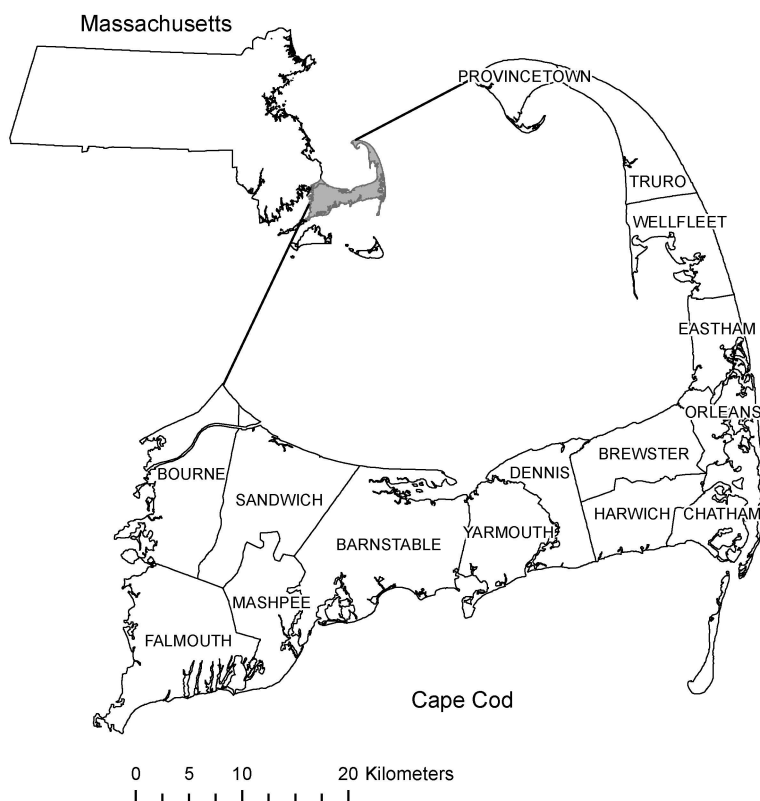


Fig. 1. Map of study area location (Cape Cod, Massachusetts, USA) showing town boundaries.

maritime climate influenced largely by the Atlantic Ocean, and consisting of average annual precipitation of about 110 cm/year characterized by warm summer (75°F) and cold winter (24°F) climatic conditions (Fletcher, 1993; Motzkin et al., 2002).

METHODS

Data

This study used the six optical bands of the Landsat-7 Enhanced Thematic Mapper-plus (ETM+) 30 m spatial resolution satellite imagery (path 11/row 31), captured on September 7, 2001. The September image was chosen because this period represented the peak in phenology, causing an improved discrimination of land-cover types. Radiometric calibration was performed on the images to convert the satellite digital numbers to at-satellite reflectance using gains and offsets obtained from the image metadata. Atmospheric correction was further employed to convert at-satellite reflectance to surface reflectance using the dark object subtraction method (Chavez, 1996). Ground reference data were collected in the field from 1025 plots having 60 × 60 m dimensions. The ground reference data were randomly split into calibration

Table 1. Number of Calibration and Validation Data for Each Land-Cover Class

Land-cover class	Number of calibration sites	Number of validation sites
Cranberry bogs	23	9
Pasture/crops	22	10
Forest	157	67
Grassland	91	33
Urban	163	69
Water	81	39
Wetland	133	65
Sand quarry	48	15

(75%) and validation (25%) subsets. The calibration and validation samples comprise 0.065% and 0.028% of the study area, respectively, and were assigned to each land-cover category as shown in Table 1. The ground reference data comprised eight different land-cover classes: cranberry bogs, pasture/row crops, forest, grassland, urban, water, wetland, and sand quarry. These map categories were determined by the Massachusetts land-cover monitoring program (MAFOMP) (Rogan et al., 2010).

Algorithm Description

Classification Tree. Classification and regression tree (CART) analysis introduced by Breiman et al. (1984) is one of the most popular non-parametric methods in land-cover classification (Hansen et al., 1996; Friedl and Brodley, 1997; Pal and Mather, 2003), and has been used in a large number of land-cover studies ranging from local to regional scales (Hansen et al., 1996; Rogan et al., 2003; Versluis and Rogan, 2010; Griffin et al., 2011). Classification trees use a hierarchical tree-based approach that partitions the measurement space recursively at the nodes until homogeneous classes are obtained based on splitting rules (Breiman et al., 1984). The splitting rules are applied to each node to determine threshold values of the splitting variables (e.g., explanatory variables, such as Landsat bands) based on the mean reduction of deviance, which increases the class homogeneity of the partitioned subset. The Gini criteria implemented in the Classification and Regression Tree (CART) package in R statistical software is used as the splitting criterion in this study. The Gini index is computed as:

$$Gini(t) = \sum_j p_j(1 - p_j)$$

where t is the node at which splitting is performed, and p_j is the relative frequency of class j at node t .

Classification trees are data-driven techniques (i.e., do not assume a particular data distribution), and are simple to interpret and operate (Rogan et al., 2003; Versluis and Rogan, 2010; Griffin et al., 2011). Additionally, they are hierarchical in nature,

and the variable used for splitting at a node can be reintroduced at a different node so that non-linearities in data relationships can be handled. Classification trees can classify remotely sensed and ancillary data that are characterized by complex measurement space, and can handle continuous as well as categorical feature variables (Fayyad and Irani, 1992; Friedl and Brodley, 1997). However, a major drawback of classification trees is that they tend to over-fit the calibration data because classification trees can generate overly complex classification rules (Prasad et al., 2006) and represent only a single iteration of the calibration data, which do not fully exploit the variation in the data (Rogan et al., 2008). Classification trees are sensitive to noisy observations and calibration data size because of their non-iterative nature (Rogan et al., 2008; Prasad et al., 2006).

Random Forests. Random forests is an ensemble learning algorithm introduced by Breiman (2001) in order to improve classification accuracies by employing multiple classifications. Random forests uses a group of tree-based classifiers $\{h(\mathbf{x}, \Theta_k), k = 1, \dots\}$ where $\mathbf{x}$ is the input vector and Θ_k is the independent and identically distributed random vectors (Breiman, 2001). A group of tree-based classifiers is generated using bootstrap samples with replacement from the calibration data. A maximum voting rule is used to assign a pixel to a class based on the maximum number of votes that the pixel receives from the group of classification trees (Ghimire et al., 2010). Each tree is trained using a certain percentage of randomly selected calibration samples with the remaining percentage of calibration samples, called out-of-bag (oob) samples, serving to estimate the classification accuracy. Additionally, the best split at each node can be obtained by a user-defined fixed number of feature/split variables (i.e., spectral bands) that can be randomly selected from the group of feature variables (Rodríguez-Galiano et al., 2011).

Random forests inherits the non-parametric nature of classification trees because it generates a group of trees to improve classification accuracy (Breiman, 2001). The random forests algorithm is readily implemented/operated because only two parameters have to be adjusted (i.e., the number of trees generated and number of feature/split variables) (Rodríguez-Galiano et al., 2012b). Furthermore, random forests does not suffer from over-fitting because the large number of trees generated ensures that the patterns in the data are generalized and can be applied to unseen samples.

Rodríguez-Galiano et al. (2012b) provided a comprehensive assessment of the random forests algorithm in land-cover classification in a complex landscape in southern Spain, and reported that the algorithm had a high overall accuracy of 92% as well as was robust to training data reduction and noise. Na et al. (2010) demonstrated that a random forests algorithm had higher accuracy than classification tree and maximum likelihood because random forests was more resistant to calibration data noise. Sesnie et al. (2010) and Pal (2005) determined that random forests and support vector machines had comparable land-cover classification accuracy but random forests required fewer user-defined parameters compared to support vector machines. Kampichler et al. (2010) compared five classification techniques for bird species mapping in the Yucatán Peninsula, and reported that random forests and classification trees performed better than artificial neural networks and support vector machines in terms of classification accuracy and classifier comprehensibility/understandability. Gislason et al. (2006) used random forests to classify forest types in a mountainous region in Colorado, and found that the random forests algorithm performed better than a

classification tree but had similar accuracy to bagging and boosting. Pino-Mejías et al. (2010) mapped oak distributions in southern Spain and determined that random forests and bagging had higher accuracy than non-ensemble learning techniques such as support vector machines, neural networks, discriminant analysis, logistic regression, and classification trees because of the input data characteristics (e.g., absence/presence training data and physiological predictor variables). However, these studies did not consider detailed multiple criteria related to calibration data and algorithm characteristics for land-cover classification and monitoring in comparing different ensemble learning algorithms.

Bagging. Bagging is an ensemble learning technique introduced by Breiman (1996) to improve classification accuracies and generalize data patterns. Similar to random forests, bagging uses a group of tree-based classifiers $\{h(\mathbf{x}, \Theta_k), k = 1, \dots\}$, where $\mathbf{x}$ is the input vector and Θ_k is the independent and identically distributed random vectors. Bagging generates a group of classification trees using a bootstrapping technique by randomly sampling certain proportions of the calibration data with replacement (Chan et al., 2001). Subsequently the multiple class assignments for each pixel generated by the classification trees is allocated to a class for which it receives a maximum number of votes from the group of classifiers (ibid.). The difference between bagging and random forests is that all the feature variables are candidates for splitting at each node, rather than using a fixed user defined number of randomly selected feature variables. Thus, bagging requires that only one parameter (i.e., the number of trees) has to be adjusted to obtain optimal classification results. DeFries and Chan (2000) found that bagging and boosting are more stable and robust to calibration data noise than classification trees, and also specify that boosting had marginally higher accuracy than bagging for land-cover classification. Similarly, Gislason et al. (2006) also found comparable accuracy between bagging and boosting in a mountainous terrain in Colorado.

Boosting. Boosting is an ensemble learning technique developed by Freund and Schapire (1996) for the purpose of improving classification accuracy by converting a group of weak classifiers (e.g., classifiers having low accuracy) to a strong classifier (e.g., classifier having high accuracy). In contrast to bagging and random forests, boosting uses the entire calibration data set for classification rather than resampling a certain proportion of the calibration data. The calibration samples are initially assigned equal weights for the first iteration of the classification and subsequently the weights are readjusted based on the fit to the calibration data in subsequent iterations of the classification tree. Misclassified calibration samples in the prior iteration are given more weight than the correctly classified calibration samples. In this way, the algorithm focuses on calibration samples that have been incorrectly classified in the prior iteration, and the classification errors that occurred in the prior iteration are corrected by the subsequent iteration. Several studies that have used boosting for land-cover categorization and monitoring using satellite data report that the technique improves land-cover classification accuracy compared to classification trees because the iterative nature of the algorithm reduces the potential for classification errors (DeFries and Chan, 2000; Muchoney et al., 2000; Friedl et al., 2002; Lawrence et al., 2004; McIver and Friedl, 2002).

Evaluation Criteria

The ground reference data and the six optical bands captured by the Landsat-7 Enhanced Thematic Mapper Plus (ETM+) instrument were input to classification tree, random forests, bagging, and boosting algorithms. Machine learning algorithms were implemented in the R statistical software (R Development Core Team, 2011) using the *rpart* package for classification trees, *ipred* package for bagging, *randomForest* package for random forests, and *adabag* package for boosting. Algorithm performance was evaluated using five criteria. The first two criteria related to the algorithm characteristics are: (1) impact of user-defined operating parameters on land-cover classification accuracy; and (2) algorithm operability and interpretability. The three criteria concerning calibration data characteristics are: (1) impact of data size; (2) data noise; and (3) data stability (i.e., calibration data variability) on land-cover classification accuracy. The map validation measures employed were overall and class-specific Kappa accuracy, which measure the agreement between the predicted and actual land-cover classes by correcting for chance agreements (Congalton, 1991). Statistical significance of differences in Kappa value calculated using four different land-cover classification methods was estimated using Z-scores (Congalton et al., 1983).

Algorithm Parameter Criteria. Ensemble learning algorithms are sensitive to the user-defined algorithm parameter values (e.g., number of trees) (DeFries and Chan, 2000). Choice of too few trees can decrease classification accuracy because of the increase in generalization error caused by the inability of the classification algorithm to generalize data patterns. Choice of an excessively large number of trees does not increase classification accuracy because the generalization error is limiting (e.g., generalization error remains constant) beyond a certain number of trees. Because bagging, boosting, and random forests use classification trees as the base classifier, the number of classification trees grown in order to operate the algorithm is the common parameter, and comparisons are performed based on the impact of number of trees on Kappa accuracy. The number of trees was varied from 1 to 1000, to evaluate the influence of number of classification trees generated on overall classification accuracy.

Algorithm Operability and Interpretability. Classification algorithms should not only provide consistent results but also should be simple to operate and interpret, which would enable the classification scheme to be readily automated in a large-area mapping context (Na et al., 2010). Automation is important in order to reduce the potential for human error caused by the analyst in performing classification, handle large cumbersome datasets encountered in mapping large complex areas, and mechanize repetitive tasks in land-cover classification (Cherrill and McClean, 1995).

Calibration Data Size. It is important to understand the influence of calibration data size on algorithm performance because the acquisition of ground reference data for classifier calibration is limited by time and cost constraints as well as scarcity of field data or historical aerial photographs for most locations (Rogan and Chen, 2004). Specifically, remote areas have limited access, and historical reference information such as air photos are not always available in repeat coverage, especially outside the United States. Furthermore, it is difficult to collect comprehensive and representative calibration data due to logistical constraints, and the large field crew required to collect data (Rogan and Chen, 2004). Thus, it is advantageous to use classification algorithms that produce high accuracy with a minimum number of ground reference samples

because of the reduction in cost and time required in field data collection. It is expected that initially, as the calibration sample size increases, classification accuracy should increase until the accuracy remains constant. The influence of calibration data size on algorithm performance was assessed by reducing the calibration data size from 100% to 10% in decrements of 10%.

Calibration Data Noise. The calibration data can be prone to noise, which can occur due to human, geometric, data format, and scale issues (Lunetta et al., 1991; Cherrill and McClean, 1995; McRoberts, 2010). Human errors can range from the analyst recording the incorrect class to changes in the land-cover conditions since field data collection and/or satellite data capture after field data collection (Lunetta et al., 1991). Data format errors are associated with calibration data capture, representation, storage, and processing in a geographic information system including geo-referencing, projection, and vector-raster conversion errors (Lunetta et al., 1991). Thus the classification algorithms employed should ideally be robust to errors in the calibration data by maintaining high classification accuracy irrespective of calibration data errors in order to improve classification accuracy (Lippitt et al., 2008). Noise in calibration data were introduced by randomly mislabeling a certain proportion of calibration classes from 0% to 90% in increments of 10%.

Stability Criterion. Understanding the influence of calibration data variations on algorithm performance is important because machine learning algorithms are often sensitive to variations in the calibration data, and calibration data are collected from multiple data sources (e.g., aerial photographs, and field-based surveys) (Lippitt et al., 2008; Rogan et al., 2008). Variations in the calibration data produce different classification rules, and change the classification accuracy, making it necessary to select classification algorithms that are insensitive to the variations in the calibration data in order to obtain consistent accuracy values. The stability criterion was assessed by randomly resampling 80% of the calibration data in 10 different iterations.

RESULTS

Algorithm Parameter Criteria

The influence of the number of trees on the overall Kappa accuracy for bagging, boosting, and random forests is shown in Figure 2. All ensemble learning algorithms showed an initial increase in Kappa accuracy of about 10% with an increase in the number of trees, but the Kappa accuracy remains constant (i.e., Kappa values between 0.75 and 0.78) as the number of trees increases beyond 50 trees. Bagging, boosting, and random forests algorithms attained a constant overall Kappa accuracy of 0.77 (Kappa range: 0.76 to 0.78), 0.78 (Kappa range 0.76 to 0.78) and 0.76 (Kappa range: 0.75 to 0.76), respectively when the number of trees was greater than or equal to 50. The overall Kappa accuracy for the ensemble learning techniques was comparable and not significantly different (Z -score < 1.96). However, bagging and random forests required fewer trees to attain a constant Kappa accuracy value.

The land-cover map produced by classification trees, bagging, boosting, and random forests as well as the locations where the ensemble learning algorithms agree and disagree is shown in Figure 3. There is similarity between classes predicted by the ensemble learning classifiers. Figure 3E displays the locations where the ensemble

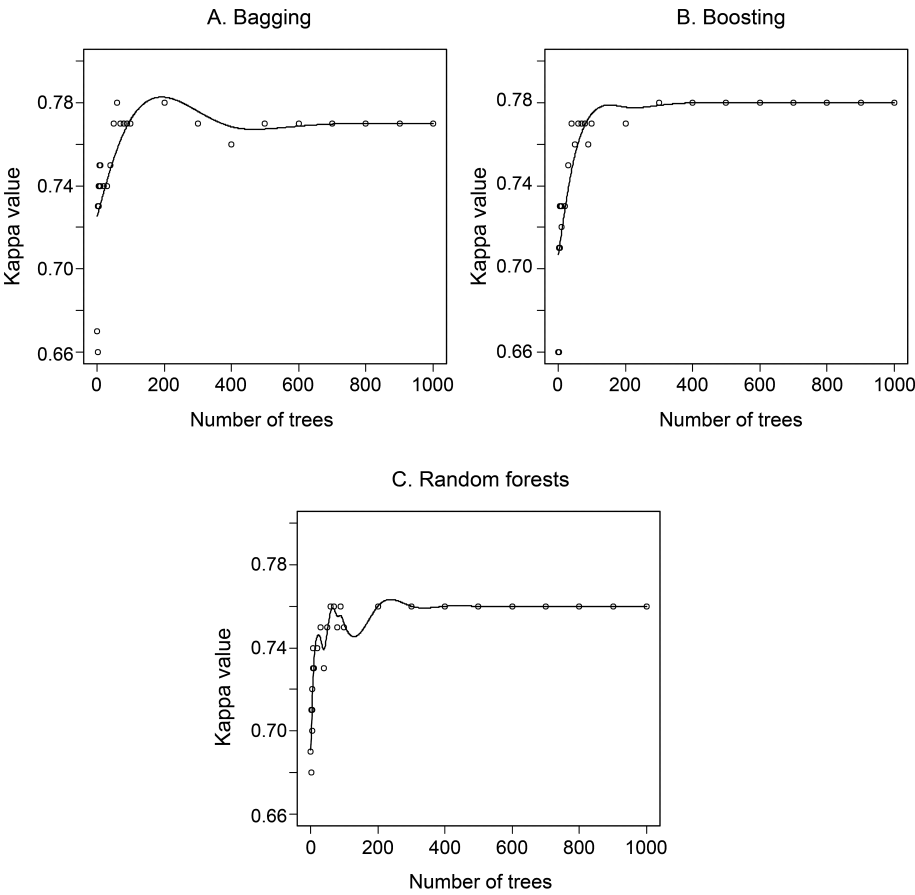


Fig. 2. Influence of the number of trees on overall Kappa accuracy value for (A) bagging, (B) boosting, and (C) random forests.

learning algorithms agree and disagree. In most of the locations, all three of the ensemble learning classifiers agree in class allocation, with disagreements being randomly distributed across the study area.

The per-class Kappa accuracy and the improvement in per-class Kappa accuracy of bagging, boosting, and random forests generated with 500 trees in comparison to the classification tree algorithm are shown in Table 2. Using classification trees, cranberry bogs and pasture/crops had the lowest accuracy. Urban and forest also had low accuracy, but were higher than the Kappa accuracy of cranberry bogs and pasture/crops. Additionally, pasture/crops had consistently low accuracy for the three ensemble learning algorithms used. Heterogeneous land-cover categories, such as cranberry bogs and forest, showed the greatest increase in per-class Kappa accuracy for bagging, boosting, and random forests. In contrast, homogeneous land-cover categories such

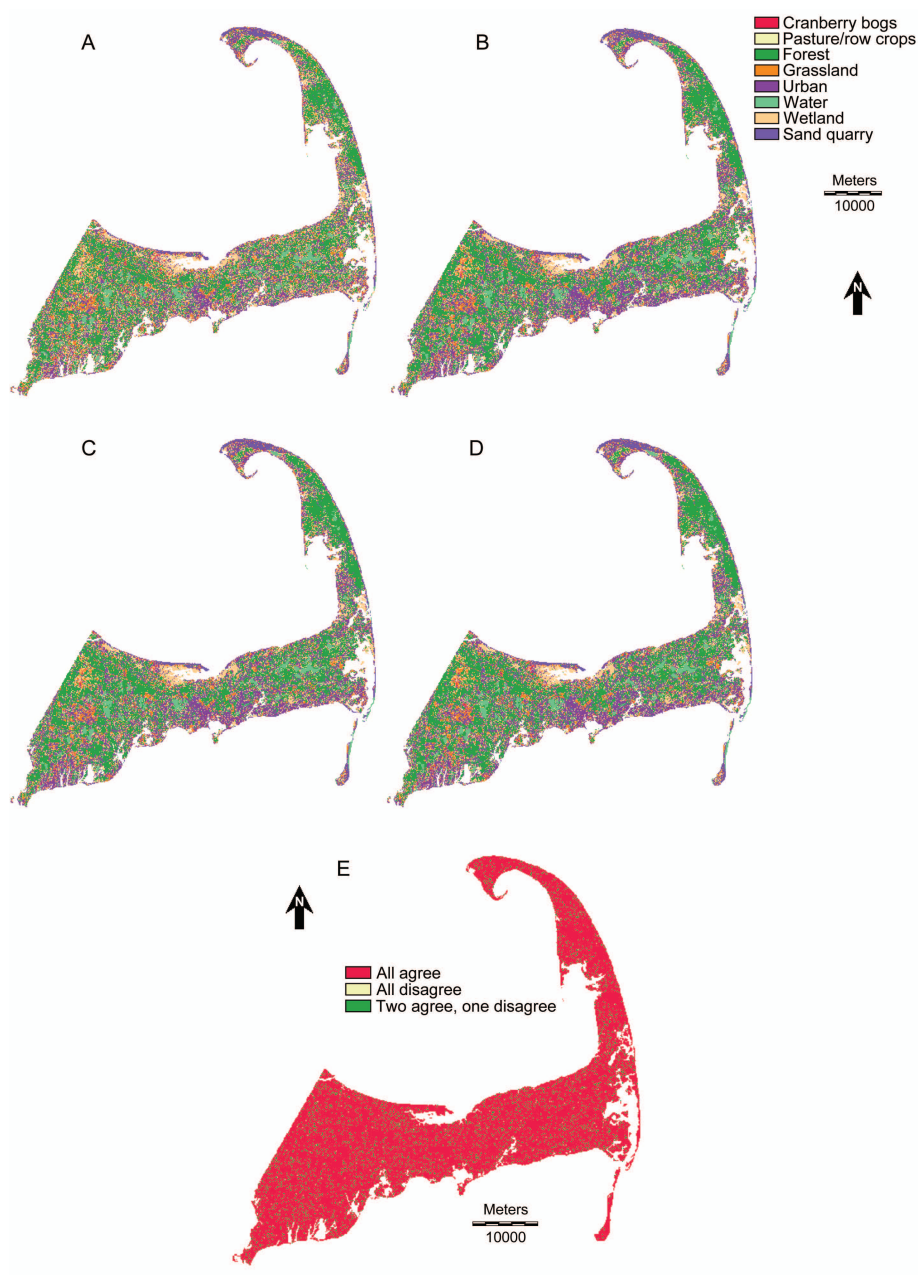


Fig. 3. Land-cover maps produced by (A) classification tree, (B) bagging, (C) boosting, and (D) random forests, and (E) map showing locations where three ensemble learning classifiers agree and disagree in terms of predicted land-cover categories.

as water did not exhibit an increase in per-class Kappa accuracy for the ensemble classifiers.

Table 2. Per-class Kappa Value and Percent Increase in Per-class Kappa for Bagging, Boosting, and Random Forests Compared to Classification Trees

Land-cover class	Classification tree	Bagging	Boosting	Random forest	Bagging increase (%)	Boosting increase (%)	Random forest increase (%)
Cranberry bogs	0.20	0.66	0.67	0.66	230	235	230
Pasture/crops	0.19	0.20	0.21	0.20	5.26	10.53	5.26
Forest	0.66	0.83	0.90	0.84	25.76	36.36	27.27
Grassland	0.72	0.75	0.82	0.72	4.17	13.89	0
Urban	0.62	0.72	0.71	0.7	16.13	14.52	12.90
Water	1.00	1.00	1.00	1.00	0	0	0
Wetland	0.71	0.73	0.73	0.73	2.82	2.82	2.82
Sand quarry	0.79	0.86	0.86	0.86	8.86	8.86	8.86

Algorithm Operability and Interpretability/Transparency

It is necessary to prune the classification tree (i.e., remove the lower splits in the classification tree) to minimize data over-fitting (Breiman et al., 1984). However, there is considerable subjectivity in creating the best-pruned classification tree (Zambon et al., 2006). In contrast, bagging, boosting, and random forests do not require pruning, and build a large number of trees that reduce generalization error and achieve higher classification accuracy (Freund and Schapire, 1996; Breiman, 2001). The process of creating a large number of trees from different subsets of the calibration data improves the ability of the algorithm to characterize the patterns in the data, and minimize the data over-fitting (Breiman, 2001). Bagging, boosting, and random forests required fewer user-defined parameters that were intuitive and easy to adjust compared to more complex algorithms (e.g., support vector machines).

Calibration Data Size

The influence of calibration data size on overall Kappa accuracy values for classification tree, bagging, boosting, and random forests is shown in Figure 4. The overall Kappa accuracy values increased by approximately 26% for classification trees and 28% for ensemble learning algorithms, with an increase in calibration data size from 10% to 100%. The lowest overall Kappa accuracy values were obtained for all the classification algorithms when 10% of calibration samples were used. Bagging, boosting, and random forests had Kappa accuracy values greater than 0.70 when at least 40% of the calibration data were used. Among the ensemble-based classification algorithms, boosting had marginally higher overall Kappa accuracy values than random forests and bagging, on average for the different calibration sample sizes. Boosting required a smaller calibration sample size of 80% to attain a level of overall Kappa

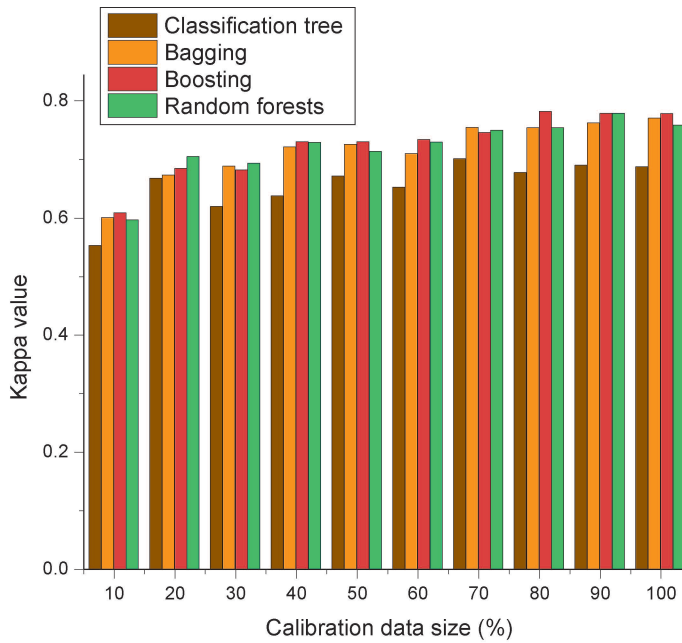


Fig. 4. Influence of the calibration data size on overall Kappa accuracy value for classification tree, bagging, boosting, and random forests.

accuracy close to the accuracy obtained by using 100% of the original calibration data. However, the overall Kappa accuracy values for bagging, boosting, and random forests were not significantly different.

Calibration Data Noise

The influence of calibration data noise on overall Kappa accuracy values for classification tree, bagging, boosting, and random forest is shown in Figure 5. Calibration data noise was categorized into three classes: low (10–20%), medium (30–60%), and high (>60%). Classification tree exhibited a linearly decreasing trend in overall Kappa accuracy with an increase in noise. Kappa accuracy values remained constant for low noise levels but decreased when the level of noise increased for the ensemble algorithms. The ensemble algorithms were robust to low to medium levels of noise, but the algorithms did not perform well at high levels of noise. All ensemble algorithms performed better than the classification tree algorithm, with Kappa increases of approximately 17–30% at low noise levels and exceeding 40% at high noise levels. At lower levels of noise, bagging, boosting, and random forests performed consistently (Kappa > 0.70), which was not much different than the case when no noise was introduced. At medium levels of noise, boosting had relatively high overall Kappa accuracy values (Kappa ~ 0.70), but there was a decrease in overall Kappa accuracy with the increase in medium levels of noise for bagging and random forests. At high noise levels all

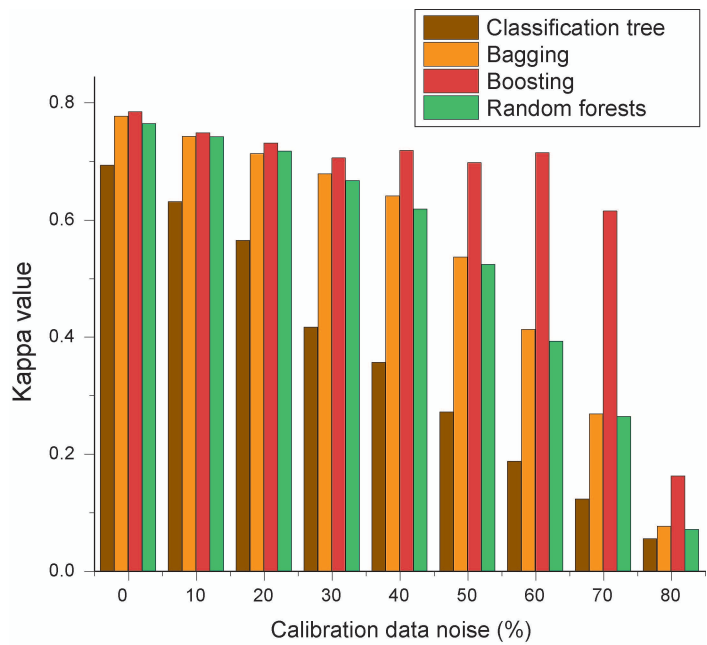


Fig. 5. Influence of the calibration data noise on overall Kappa accuracy value for classification tree, bagging, boosting, and random forests.

the ensemble learning algorithms demonstrated a decrease in overall Kappa accuracy, with the increase in the level of noise with boosting being least affected by noise.

Stability Criterion

The influence of calibration data variations on the overall Kappa accuracy values is shown in Figure 6. All the algorithms were not influenced by variations in the calibration data because of low standard deviations (standard deviation < 0.02) in Kappa accuracy for all algorithms with calibration data variations. Bagging, boosting, and random forests performed consistently without appreciable differences in overall Kappa accuracy. Combined samples for the ensemble algorithms had a mean overall Kappa accuracy value of 0.77. Among the ensemble learning algorithms, bagging had the highest overall Kappa accuracy range (Kappa range: 0.73–0.79) followed by boosting (Kappa range: 0.75–0.80) and random forests (Kappa range: 0.75–0.79). In contrast, the classification tree had a relatively lower mean overall Kappa accuracy value of 0.69, with a standard deviation of 0.02 and range of approximately 0.66 to 0.72. However, the classification tree had a comparable and low standard deviation, inferring that the classification tree was equally insensitive to variations in the calibration data. Ensemble learning techniques and classification trees were robust to variations in different random subsets of the calibration data because of the low standard deviations of the accuracy values.

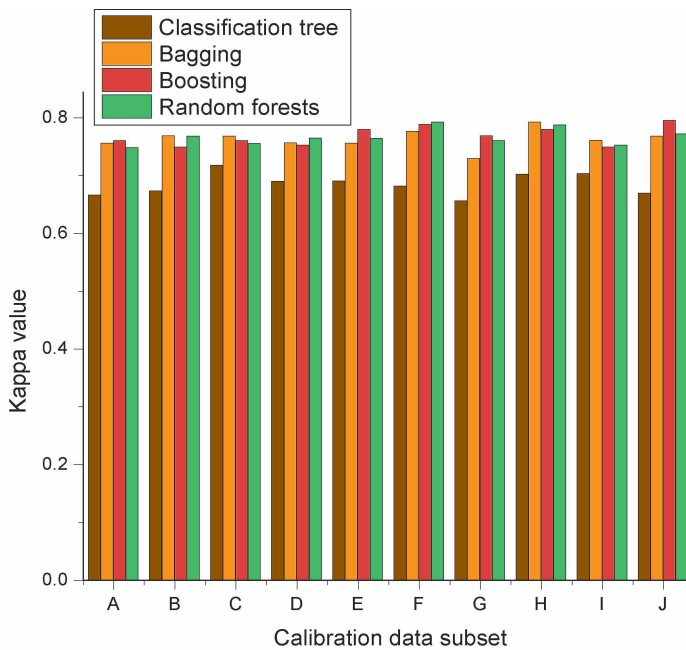


Fig. 6. Influence of the calibration data variations on overall Kappa accuracy value for classification tree, bagging, boosting, and random forests.

DISCUSSION AND CONCLUSIONS

The purpose of this study was to evaluate the performance of classification tree and ensemble learning algorithms (bagging, boosting, and random forests) in a heterogeneous landscape. Five different criteria were used in the evaluation, out of which two criteria are related to user-defined algorithm operating parameters and algorithm interpretability, and the other three criteria assess the effects of calibration data size, noise, and the stability of land-cover classification accuracy. The ensemble learning techniques outperformed the classification tree algorithm in terms of land-cover classification accuracy, especially for heterogeneous categories which occupied small areas, varied seasonally, or were comprised of several subcategories. Boosting was most robust to low and medium levels of calibration data noise (Kappa accuracy > 0.70), followed by random forests and bagging. All ensemble algorithms were comparably robust to low and medium reductions in calibration data size (Kappa accuracy > 0.70) and insensitive to variability in calibration data. Calibration data noise was observed to be the most important factor influencing the accuracy, because high levels of noise substantially reduced Kappa accuracy by more than 40% in most instances. This study suggests that ensemble learning techniques can outperform non-ensemble classifiers in heterogeneous landscapes by better handling calibration data noise and small calibration data sizes.

Bagging and random forest algorithms performed similarly, in terms of the number of trees used to operate the algorithms, because both bagging and random forests

use bootstrap sampling techniques on the calibration data to generate an ensemble of trees. The land-cover class for each pixel in an image is based on a majority voting rule by the ensemble of trees. However, bagging and random forests differ in terms of variable selection. The random forests algorithm randomly selects a user-defined number of potential splitting variables at each node, whereas the bagging algorithm uses all the potential splitting variables. Thus, a random forests algorithm in which all of the splitting variables are used can be considered as a bagging algorithm. In contrast to the random sampling of calibration data and feature variables employed by bagging and random forests, the boosting algorithm uses a conceptually different technique. The boosting algorithm is based on iteratively classifying data by giving more weight to pixels misclassified in the prior iteration (Freund and Schapire, 1996). The conceptual dissimilarity of the boosting algorithm to bagging and random forests caused the boosting method to perform marginally differently, but at a comparable level to bagging and random forests in terms of the number of trees required to attain a constant accuracy value.

Classification trees suffer from data overfitting, which occurs because instead of generalizing the patterns in the data, classification trees can be prone to noise in the calibration data. In contrast, ensemble learning algorithms can generate a large number of trees for classification, which ensures that the learning algorithm can generalize the patterns in the data (Breiman, 2001). Choice of the number of trees is intuitive because usually a large number of trees (>50–100) produces high accuracy. Among the ensemble learning techniques, boosting was more robust to calibration data noise as well as marginally better in terms of calibration data size compared to bagging and random forests. This suggests that boosting would be suitable in a large-area mapping project where cost and time constrain the collection of a large number of high quality calibration samples because it is less sensitive to calibration data size and noise than bagging and random forests. Heterogeneous land-cover categories (such as cranberry bogs and forest) are difficult to classify because of seasonal variability and lumping of several finer subcategories into one broad category. The iterative nature of the ensemble learning algorithms was able to increase the accuracy of heterogeneous land-cover categories (e.g., cranberry bogs, forest, and pasture/crops) by improved generalization of relations and patterns between the calibration data and spectral bands. This increase in per-class accuracy of heterogeneous land-cover categories resulted in improved overall accuracy of the ensemble learning techniques compared to classification tree. These heterogeneous categories benefited the most by the use of ensemble learning algorithms because these categories had a greater increase in accuracy.

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